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Sem Start: 05 Jan 2026	Sem End: 18 April 2026

Assignment No	Description	USN	Name	Due	Remarks

Formwork Selection for a 30-Storey Residential Building

Site: 30m × 30m | Residential High-Rise

1. Introduction

This assignment evaluates formwork systems for a 30-storey residential building on a constrained 30×30m site. The selected formwork must optimize cost, speed, repetition, labour efficiency, and safety while accommodating an innovative facade system. After comparative analysis, **Mivan (Aluminium) Formwork** is selected as the most suitable system.

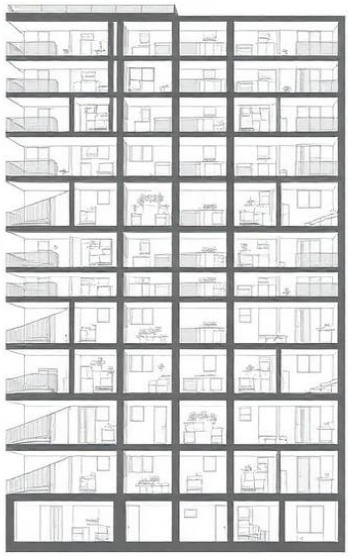

Fig 1: Typical section of a high-rise residential building

2. Project Parameters

Parameter	Detail
Building Type	30-Storey Residential Tower
Site Area	30m × 30m (900 sq.m)
Typical Floor Height	3.0m floor-to-floor
Structure	RCC frame with shear walls
Floor Plate	Repetitive layout across 28 typical floors
Facade	Prefabricated Insulated Composite Panels (PICP)

3. Selected Formwork: Mivan (Aluminium) Formwork

Mivan formwork, originally developed in Europe and commercialised by MIVAN Company Ltd (Malaysia) in the 1990s, uses precision-engineered aluminium alloy panels to cast walls and slabs monolithically. It is the industry standard for mass housing and high-rise residential projects with repetitive floor plans.

Fig 2: Mivan aluminium formwork panels on a high-rise construction site

3.1 Key Components

The Mivan system comprises wall panels, deck panels (for slabs), props, rockers, stiffeners, kickers, and pin connectors. All components are fabricated from high-grade aluminium alloy (typically 65ST6), making them lightweight yet structurally robust.

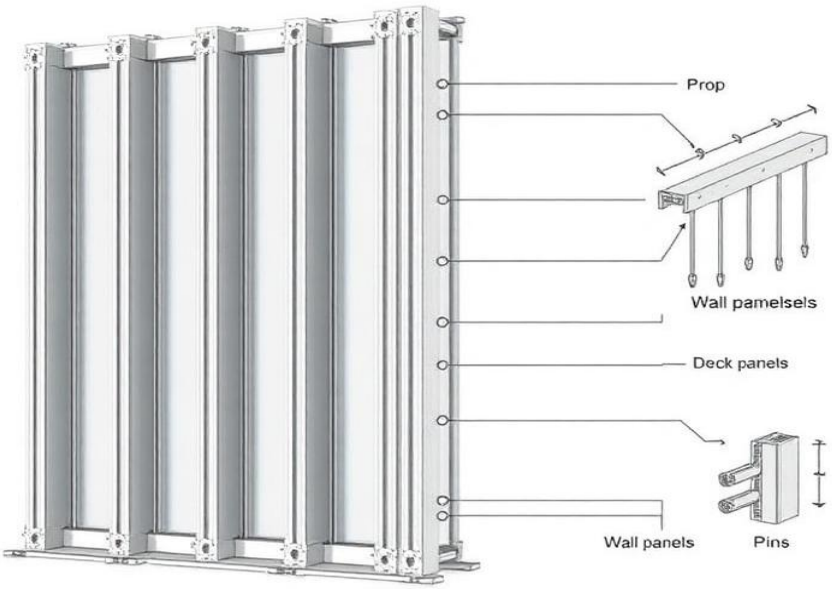

Fig 3: Components of Mivan formwork system — wall panels, deck panels, props, and pins

3.2 Justification for Selection

Cost

Initial investment for Mivan is higher than conventional formwork. However, the aluminium panels can be reused over 200–250 times (Nilkanth Aluform, 2025). For a 30-storey building with 28 typical floors, the cost per floor decreases significantly after the first few cycles. Overall project cost reduces by 15–20% compared to conventional methods when repetition exceeds 15 floors (JK Cement, 2025).

Speed

Mivan enables a floor cycle time of 7–10 days per floor, compared to 15–21 days for conventional formwork. For 30 storeys, this translates to a potential saving of 5–7 months on the structural programme. The monolithic casting of walls and slabs eliminates the need for separate wall and slab operations (Nobroker, 2025).

Repetition

Residential towers have highly repetitive floor plates. With 28 identical typical floors on a 30×30m site, the same set of Mivan panels is reused on every floor without modification. This is where Mivan outperforms all other systems — the formwork is custom-designed once and repeated throughout (Homes247, 2025).

Labour

Mivan requires fewer skilled workers compared to conventional formwork. A typical floor requires 20–25 workers for assembly and stripping, versus 40–50 for conventional methods. The system uses semi-skilled labour after initial training, reducing dependence on specialised carpenters (RE Consult Advisory, 2025).

Safety

Aluminium panels are lightweight (18–22 kg/m²) and can be handled manually without cranes for most operations. Built-in platforms, guardrails, and walkway brackets are part of the system. The monolithic pour eliminates loose scaffolding and temporary props at height, significantly reducing fall hazards (BFS Industries, 2026).

4. Comparison with Other Formwork Systems

Criteria	Mivan	Conventional	Climbing	Slip	Table/Flying
Cost (per floor)	Low (after 10+ reuses)	High (recurring)	Moderate	Moderate-High	Moderate
Speed (days/floor)	7–10	15–21	10–14	Continuous but limited	10–14
Repetition Suitability	Excellent (200+ reuses)	Poor	Good for core walls	Good for shafts only	Good for slabs only
Labour Required	20–25	40–50	15–20	10–15 (specialised)	15–20
Safety	High (built-in platforms)	Low (temporary scaffolding)	High (enclosed platforms)	Moderate (continuous operation)	Moderate
Facade Compatibility	Excellent	Good	Limited (obstructs facade)	Poor (no stops allowed)	Fair (slab edge only)

5. Why Other Systems Are Less Suitable

Conventional Formwork

Timber and steel conventional formwork is labour-intensive, slow (15–21 days/floor), and produces inconsistent surface finishes. For 30 storeys, the cumulative time and labour cost makes it economically unviable. Timber panels degrade after 8–12 uses, generating waste. It also cannot achieve the dimensional precision needed for prefabricated facade panel integration (BFS Industries, 2026).

Climbing Formwork

Climbing formwork (hydraulic or crane-assisted) is suited for core walls, lift shafts, and non-repetitive vertical elements. It cannot cast floor slabs, requiring a separate slab formwork system. For a residential building where walls and slabs need simultaneous casting, climbing formwork alone is insufficient. It also obstructs facade installation on the levels where the climbing frame is positioned (Rapid Scafform, 2026).

Slip Formwork

Slip formwork provides continuous vertical casting at 150–300mm per hour. It is ideal for chimneys, silos, and bridge piers but not for residential floors with openings (windows, doors, service ducts). Stopping and restarting the slip causes cold joints. The PICP facade system requires precise anchor points at each floor, which a continuously moving formwork cannot accommodate (Libmatt, 2025).

Table / Flying Formwork

Table formwork (flying forms) is efficient for flat slab construction but handles only horizontal elements. Walls must be cast separately using another system. For a 30-storey residential building where wall-slab monolithic casting is desired, table formwork is only a partial solution and would require combining with another wall formwork system, increasing complexity and cost (Orion PPM, 2026).

6. Innovative Facade System: Prefabricated Insulated Composite Panels (PICP)

The proposed facade enclosure uses **Prefabricated Insulated Composite Panels (PICP)** — a system where factory-made panels consisting of an outer aluminium composite sheet, rigid polyisocyanurate (PIR) insulation core, and inner fibre-cement board are bolted directly onto cast-in-situ anchor brackets at each floor level.

Fig 4: Prefabricated facade panels being installed on a high-rise building

6.1 Why PICP Works with Mivan

Mivan formwork produces walls with dimensional tolerance of ±2mm. This precision is essential for PICP panels, which are factory-manufactured to exact dimensions. The cast-in anchor brackets are positioned within the Mivan panel layout before the concrete pour, ensuring perfect alignment floor after floor. No post-casting drilling or modification is needed.

6.2 Why Other Formworks Fail with PICP

System	Issue with PICP Facade
Conventional	Surface irregularities (±10–15mm) make panel alignment difficult; requires extensive plastering and preparation b
Climbing	Climbing frame occupies the facade zone during casting; panels cannot be installed until formwork is moved up,
Slip	Continuous movement prevents embedding anchor brackets at precise levels; cold joints from stopping comprom
Table/Flying	Does not cast walls; wall casting by another method introduces tolerance mismatches between wall and slab edg

7. Conclusion

For a 30-storey residential building on a constrained 30×30m site, **Mivan aluminium formwork** is the optimal choice. Its ability to cast walls and slabs monolithically in 7–10 day cycles, combined with 200+ panel reuses across 28 repetitive floors, delivers unmatched speed and cost efficiency. The dimensional precision of ±2mm enables seamless integration with the proposed PICP facade system. Other formwork systems either lack speed (conventional), cannot cast slabs (climbing), cannot accommodate openings (slip), or only handle slabs (table formwork) — making them less suitable for this project.

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